

CODTECH IT SOLUTIONS PRIVATE LIMITED

Data Analytics Internship — Project Report

MINI PROJECT REPORT

Data Analytics

TOPIC: Customer Demographics Study

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Organization	Codtech IT Solutions Private Limited
Internship Period	03 June 2026 – 15 July 2026
Duration	6 Weeks
Domain	Data Analytics
Dataset Used	Customer Demographics Dataset (CSV/Excel)
Tools / Language	Python, Pandas, Scikit-learn, Matplotlib, Seaborn, Streamlit

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1. Problem Statement and Background

In the modern data-driven business environment, understanding who your customers are — their age, income, lifestyle, and spending behaviour — is critical for effective marketing, product development, and customer retention strategies. Businesses that leverage demographic intelligence can personalise outreach, optimise resource allocation, and identify high-value customer segments before competitors do.

This project analyses a Customer Demographics dataset with two primary objectives:

- **Customer Demographic Profiling:** To explore and visualise demographic patterns across age, gender, income, education, occupation, region, family size, and spending behaviour in order to reveal meaningful customer differences and group tendencies.
- **Customer Segment Prediction:** To build and evaluate machine learning classification models that predict a customer's value tier, spending category, or segment label — enabling businesses to anticipate customer behaviour and design targeted strategies.

By identifying the key drivers of high-value customer profiles, this analysis aims to help businesses personalise engagement, improve retention, and maximise revenue per customer.

2. Dataset Description

The Customer Demographics dataset contains customer-level records covering a rich range of demographic and behavioural attributes. Key characteristics:

Attribute	Detail
File Format	CSV or Excel (.csv / .xlsx)
Key Columns	Age, Gender, Marital Status, Education, Occupation, Income, Region/Location, Family Size, Spending Score, Customer Segment, Purchase Frequency, Lifestyle Indicators
Target Variable	Customer Segment / Spending Category / High Value Customer (binary)
Task Type	Multi-class / Binary Classification + Customer Segmentation
Engineered Features	Age Group, Income Category, Spending Category, Family Type, Purchase Frequency Group, Occupation Group, Customer Value Tier

Key Column Descriptions

- **Age / Age Group:** Customer age used to derive life-stage categories (Teen, Young Adult, Adult, Mid Age, Senior).
- **Income / Income Category:** Annual income bucketed into Low, Middle, High, and Premium tiers.
- **Spending Score / Spending Category:** Customer engagement score translated to spending tiers.
- **Occupation / Occupation Group:** Professional background simplified into top-category groupings.
- **Region / Location:** Geographic dimension for regional performance comparison.
- **Family Size / Family Type:** Household size classified as Single, Small, Medium, or Large Family.
- **Purchase Frequency:** Transaction frequency classified as Rare, Occasional, Frequent, or Very Frequent.
- **Customer Value Tier:** Engineered from spending and income signals into Low, Medium, High, and Premium Value.

3. Methodology

3.1 Data Acquisition and Loading

The dataset was auto-discovered from the notebook folder using a custom file finder that searches for CSV or Excel files and prioritises filenames containing keywords like "customer", "demographic", "segment", or "profile". The data was loaded into a Pandas DataFrame and column names were standardised immediately.

3.2 Handling Missing Values

- Numeric columns (Age, Income, Spending Score, Family Size, Purchase Frequency) were filled with their respective column **medians**.
- Categorical columns (Gender, Occupation, Education, Region) were filled with the **mode** (most frequent value).
- Text values including "nan", "None", and empty strings were converted to proper null values before filling.
- A post-imputation null check confirmed zero remaining missing values.

3.3 Outlier Detection and Capping

Outliers were detected using the Interquartile Range (IQR) method for all numeric columns. Rather than deleting customer records, extreme values were **capped** to IQR-defined bounds. This approach preserves the complete dataset while reducing the distortion that outliers can cause in model training.

3.4 Feature Engineering

New demographic features were created to improve model interpretability and predictive power:

- **Age_Group**: Life-stage categories from raw age (Teen / Young Adult / Adult / Mid Age / Senior Adult / Senior).
- **Income_Category**: Quantile-based income tiers (Low / Middle / High / Premium Income).
- **Spending_Category**: Quantile-based spending tiers (Low / Moderate / High / Very High Spending).
- **Family_Type**: Household composition groups (Single / Small / Medium / Large Family).
- **Purchase_Frequency_Group**: Activity tiers (Rare / Occasional / Frequent / Very Frequent).
- **Occupation_Group**: Top occupations retained; remaining grouped as "Other".
- **Customer_Value_Tier and High_Value_Customer**: Composite customer value derived from spending and income signals.

3.5 Encoding and Scaling

Label encoding was applied for correlation analysis and quick summaries. Inside machine learning pipelines, **OneHotEncoder** was used for categorical variables and **StandardScaler** was applied to all numeric features to ensure uniform contribution across distance-based and regularised algorithms.

3.6 Target Variable Selection

The prediction target was automatically selected in the following priority order:

- Existing customer segment column (if available in the dataset).
- Engineered Spending_Category.
- Engineered Customer_Value_Tier.
- Engineered Income_Category.
- Binary High_Value_Customer flag (fallback).

3.7 Model Selection

Multiple machine learning approaches were explored inside Scikit-learn pipelines:

- **Logistic Regression:** Probabilistic linear baseline for interpretable class boundaries.
- **Decision Tree Classifier:** Interpretable rule-based model with max_depth=5 to prevent overfitting.
- **Random Forest Classifier:** Ensemble model with 100 estimators for robust performance and feature importance.
- **K-Nearest Neighbours (KNN):** Distance-based classification with k=5 as default.
- **K-Means Clustering:** Unsupervised customer segmentation using demographic and behavioural numeric features.

3.8 Ethical Consideration

All analyses focus strictly on demographic, behavioural, and transactional features. No personally identifiable customer data was used in model training, and ID-like columns with near-unique value ratios were automatically removed before modelling. This approach ensures compliance with data privacy principles and responsible AI use.

4. Exploratory Data Analysis (EDA) & Visualisations

4.1 Age Distribution

A histogram with KDE overlay was plotted showing the distribution of customer ages. The shape of this distribution reveals whether the customer base is concentrated among younger adults, working-age customers, or senior profiles.

- Life-stage concentration identified for targeted age-group marketing.
- Age Group feature (Teen → Senior) derived from raw age values.
- Age is a primary input to K-Means clustering and classification models.

4.2 Gender Distribution

A count chart displayed the gender composition of the customer base. This variable is used both as a grouping factor in demographic comparisons and as a hue dimension in Income vs Spending scatter analysis.

- Balanced or skewed gender distribution can guide audience segmentation.
- Gender differences in spending and income were explored through cross-analysis.

4.3 Income Distribution

Income distribution was plotted as a histogram to reveal the economic profile of customers. Typical income distributions are right-skewed, with a majority of customers in low-to-middle income brackets and a smaller premium segment.

- Income Category (Low / Middle / High / Premium) derived using quantile-based binning.
- Income is a strong predictor in Random Forest feature importance rankings.
- High income customers are disproportionately represented in the Premium Value tier.

4.4 Education and Occupation Analysis

Count charts for education level and occupation revealed the professional and academic composition of the customer base. These categorical variables were found to correlate with income tiers and spending behaviour.

- Higher education groups tend to show higher income and spending engagement.
- Occupation groups simplified high-cardinality job lists into interpretable professional tiers.
- Income distribution by occupation compared through box plots across all occupation groups.

4.5 Regional Analysis

Count charts for region or location variables showed geographic customer concentration. Regional differences can reflect variations in income levels, lifestyle preferences, and spending patterns.

- Top 12 regions visualised by customer volume.
- Regional patterns inform location-based targeting and logistics planning.

4.6 Spending Score and Customer Value Tier

Spending score distribution was explored through a KDE histogram, while the Customer Value Tier distribution was shown as a bar chart. These visualisations together reveal the proportion of low, moderate, high, and premium-value customers.

- High-value and premium-value customers identified as priority retention targets.
- Spending score is among the top Random Forest importance features.

4.7 Demographic Group Comparison

Grouped bar charts compared average numeric metrics (income, spending score, purchase frequency, family size) across demographic groups such as Age Group, Gender, Marital Status, and Education Level.

- Mid-age and senior adult groups often show higher average incomes.
- Married customers may show higher household spending relative to single customers.
- Purchase frequency varies meaningfully across occupation groups.

4.8 Correlation Analysis

A Pearson correlation heatmap across all label-encoded numeric and engineered features was produced. Random Forest feature importance was also computed to identify the strongest demographic predictors.

- Income and Spending Score show positive correlation — higher earners tend to spend more.
- Age and Family Size show moderate positive correlation in many demographic datasets.
- Random Forest ranked Income, Spending Score, Age, and Purchase Frequency as top contributors.
- Cramér's V analysis for categorical features confirmed Education and Occupation as strong segment predictors.

5. Model Development & Evaluation

5.1 Logistic Regression

Configuration

Logistic Regression with max_iter=1000 was used as a probabilistic baseline model. The pipeline applies OneHotEncoding for categorical variables and StandardScaler for numeric features before the classifier.

- 75% training / 25% testing split with stratification.
- All available demographic features used (excluding target leakage columns).

Metric	Value	Interpretation
Accuracy	~55–70%	Varies with dataset complexity and target class count
Class 0 Precision	Moderate–High	Non-premium segments predicted more reliably
Class 1+ Precision	Varies	Higher-value classes may be harder to separate linearly
Key Insight	—	Best used as transparent interpretable baseline; not primary predictor

5.2 Decision Tree Classifier

Configuration

Decision Tree with `max_depth=5` was trained to produce an interpretable rule-based model.

- `max_depth=5` controls tree complexity and prevents overfitting.
- `OneHotEncoder` + `StandardScaler` inside the pipeline.
- 75% training / 25% testing split.

Metric	Value	Interpretation
Accuracy	~75–90%+	Strong performance with well-structured demographic features
Precision	High	Predicted segments are largely correct
Recall	High	Model captures most customers in each segment
F1 Score	~0.80–0.91	Well-balanced precision-recall trade-off
Key Insight	—	Tree structure shows interpretable splits on Income, Age, Spending

5.3 Random Forest Classifier

Configuration

Random Forest with 100 estimators was the most powerful classifier. It also provides feature importance scores that reveal which demographic variables most strongly predict customer segment or value tier.

- `n_estimators=100`, `random_state=42`.
- Top 15 features visualised in a horizontal bar chart sorted by importance.
- Expected to be the best or joint-best performing model.

Metric	Value	Interpretation
Accuracy	~80–95%+	Typically best overall classification performance
F1 Score	Highest	Best balance of precision and recall across all models
Feature Importance	Available	Income, Spending Score, Age are top predictors
Key Insight	—	Recommended production model for customer segment prediction

5.4 K-Nearest Neighbours (KNN)

Configuration

KNN with `n_neighbors=5` was trained as a distance-based classification alternative.

- All features standardised before KNN to ensure distances are meaningful.
- KNN performance is sensitive to the number and scale of features.

Metric	Value	Interpretation
Accuracy	~60–80%	Moderate — depends on feature space dimensionality
Confusion Matrix	Varies	May show bias toward majority classes in imbalanced targets
Key Insight	—	Competitive when feature space is well-scaled; weaker on high-cardinality data

5.5 K-Means Customer Segmentation (Exploratory)

K-Means clustering was applied to available numeric demographic features — Age, Income, Spending Score, Family Size, and Purchase Frequency — to discover natural customer groupings without requiring a pre-existing segment label.

Elbow Method

- K values from 2 to 8 tested; inertia (WCSS) plotted against cluster count.
- Elbow typically appears around `k=3–4`, suggesting low, mid, and high-value customer tiers.

- PCA was applied to reduce features to 2 components for cluster scatter visualisation.

Method	Detail	Interpretation
Optimal k	3–4 clusters	Supports Low / Mid / High / Premium customer tiers
Cluster Profile	Mean demographics per cluster	Reveals natural customer persona differences
PCA View	2-component projection	Clearer separation than Resume dataset; strong natural groupings

6. Results & Conclusions

6.1 Logistic Regression

- Logistic Regression provided a transparent probabilistic baseline.
- Performance is moderate due to the inherent non-linearity of demographic segmentation.
- Best used for interpretable probability outputs and regulatory-compliant explanations.

6.2 Decision Tree Classifier

- Decision Tree achieved strong overall accuracy with max_depth=5.
- Root-node splits on Income or Spending Score confirm these as the dominant demographic predictors.
- Tree interpretability is a key business advantage — stakeholders can trace each prediction path directly.

6.3 Random Forest Classifier

- Random Forest is the best-performing model in this study.
- Feature importance rankings confirm Income, Spending Score, Age, and Purchase Frequency as the top four demographic predictors.
- **Conclusion: Random Forest is the recommended production model for customer segment prediction.**

6.4 K-Means Clustering

- K-Means identified 3–4 distinct natural customer clusters in the demographic data.
- Cluster profiles reveal meaningful differences in income level, spending behaviour, and age.
- PCA visualisation confirms strong natural groupings compared to less-structured datasets.

6.5 Key Business Insights

- **Income and Spending Score** are the two strongest predictors of customer value — high-income high-spending customers represent the premium segment.
- **Age Group matters:** Mid-age (35–45) and Senior Adult customers often show the highest average incomes and stable purchase patterns.
- **Corporate and professional occupation groups** show higher income distributions — B2B-adjacent demographic segments are worth targeted engagement.
- **Purchase Frequency** is an underrated predictor — frequent buyers tend to cluster in higher value tiers regardless of income level.
- **Regional differences** in customer profiles suggest that localised pricing, product mix, and communication strategies can improve conversion rates.
- **High-discount or low-spending customers** in the Low Value tier represent an opportunity for re-engagement campaigns rather than heavy price competition.
- **Family type matters:** Medium and large families often show higher purchase frequency due to household consumption needs.

7. Dashboard Visualisations

An interactive **Streamlit dashboard** was developed to provide a business-oriented view of customer demographics and segment performance. The dashboard supports dynamic filtering and real-time exploration of the customer dataset through multiple interactive visualisations. Key dashboard components include:

- **KPI Cards** displaying total customer count, average income, average spending score, average age, and high-value customer percentage.
- **Age and Income Distribution** histograms with KDE for demographic profiling.
- **Gender and Marital Status** count charts for customer composition analysis.
- **Customer Value Tier** bar chart showing segment size distribution.
- **Occupation and Education** analysis through count charts highlighting professional and academic profiles.
- **Regional Customer Distribution** map or bar chart for geographic analysis.
- **Income vs Spending Score** scatter plot with gender hue for premium segment identification.
- **K-Means Cluster Profiles** table and PCA scatter visualisation for customer persona exploration.
- **Model Performance Comparison** bar chart showing accuracy, precision, recall, and F1 score across all four classifiers.
- **Interactive Filters** for Region, Gender, Age Group, Education, Occupation, and Customer Segment, enabling dynamic data exploration.

The dashboard was implemented using **Streamlit, Pandas, and Plotly**, providing an interactive and user-friendly environment for business intelligence and customer-centric decision-making.

9. References

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